Notes: Schroth, Suarez, Taylor (JFE, 2014)

Dynamic Debt Runs and Financial Fragility: Evidence from the 2007 ABCP Crisis

Contents

1 What the paper is doing

The paper estimates a structural *dynamic debt-run* model on the 2007 U.S. asset-backed commercial paper (ABCP) crisis, emphasizing how **endogenous rollover yields create dilution risk**, and then uses the estimated model to quantify fragility, run warning signals, and policy-relevant sensitivities.

2 Institutional setting and empirical objects

2.1 ABCP conduits and runs

An ABCP conduit is a short-term-financed vehicle that holds a portfolio of (mostly) fixed-income assets and funds them with very short-term paper. The 2007 ABCP crisis is a good laboratory because:

- yield spreads rose sharply before runs,
- runs happened at the *program* level (conduit-by-week), and
- conduits differed in guarantee strength, generating useful cross-sectional variation.

2.2 Data and key definitions

The dataset (from DTCC) covers essentially all U.S. ABCP issuance transactions in 2007 and allows weekly program-level outstanding amounts and yields.

Guarantee types and subsamples. Using Moody's classifications, conduits fall into four broad guarantee categories (full credit, full liquidity, SIV, and extendible notes). The empirical strategy splits the sample into:

- **Strong guarantees**: full credit or full liquidity (191 conduits), 45% in a run during 2007;
- **Weak guarantees**: SIV or extendible notes (90 conduits), 83% in a run during 2007.

All structural parameters are assumed constant *within* each subsample.

Run indicator and recovery. A conduit i is "in a run" in week t if either:

1. more than 10% of its outstanding paper is scheduled to mature but the conduit issues *no* new paper that week; or
2. it was in a run in week $t-1$ and issues no new paper in week t.

A conduit *recovers* in week t if it issues paper in week t after being in a run in week $t-1$.

Rollover yield spread. The weekly rollover yield spread is the dollar-weighted annualized yield on paper issued (using Thursday issuance, or the next issuance day if no Thursday issuance) minus the fed funds rate.

3 Core model primitives

3.1 Time, agents, and risk-neutral pricing

Time is continuous. Agents are risk-neutral with discount rate ρ (interpretable as the risk-free rate).

3.2 Asset payoff, random maturity, and fair value

The conduit holds an asset that pays out at a random maturity time τ_ϕ that arrives with Poisson intensity ϕ. The state variable y_t follows geometric Brownian motion:

$$\frac{\mathrm{d}y_t}{y_t} = \mu \,\mathrm{d}t + \sigma \,\mathrm{d}Z_t. \tag{1}$$

Hence the asset's time-t value is

$$F(y_t) \equiv \mathbb{E}_t\left[e^{-\rho(\tau_\phi - t)} y_{\tau_\phi}\right] = \frac{\phi}{\rho + \phi - \mu}\, y_t. \tag{2}$$

3.3 Debt rollover: Calvo-style maturity and debt growth

The conduit finances the asset with short-term zero-coupon debt. Individual debt matures with intensity δ, so average maturity is $1/\delta$.

Let R_t be the face value promised per dollar rolled over at time t. Let D_t be total face value of debt outstanding. Since a fraction $\delta\,\mathrm{d}t$ matures each instant, total face value evolves as

$$\mathrm{d}D_t = \delta D_t (R_t - 1)\,\mathrm{d}t. \tag{3}$$

When $R_t > 1$, rollover increases promised face value over time.

3.4 Runs, liquidation, and sponsor guarantees

A run occurs when maturing creditors refuse to roll over. If funding cannot be rolled over and the sponsor backstop fails, the conduit liquidates at a fire-sale discount:

$$L(y_t) \equiv \alpha F(y_t), \qquad \alpha \in (0, 1].$$

Higher α means more liquid assets and higher recovery.

Sponsors provide a backstop (credit line / liquidity guarantee). Once a run begins, maturing paper can be paid by borrowing from the sponsor, but the credit line fails with probability $\theta\delta\,\mathrm{d}t$ each instant. Thus expected guarantee survival time is $1/(\theta\delta)$, and **higher θ means weaker guarantees**.

3.5 Yield spreads and a cap

Let r_t denote the annualized yield spread above the risk-free rate. Face values map to spreads via

$$r_t = (R_t - 1)(\phi + \delta) - \rho. \tag{4}$$

A key institutional feature is that rollover yields cannot rise without bound (e.g. money market fund constraints, adverse selection/credit rationing). The model imposes a cap on rollover yields: either an upper bound $\bar{R}$ on face value or an upper bound $\bar{r}$ on spreads.

3.6 State variable: inverse leverage

Define inverse leverage

$$x_t \equiv \frac{y_t}{D_t}.$$

Using Itô and (3),

$$\frac{\mathrm{d}x_t}{x_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t + \delta\,\mathrm{d}t - \delta R_t\,\mathrm{d}t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t - \delta(R_t - 1)\,\mathrm{d}t. \tag{5}$$

This is the key feedback channel: when R_t rises, debt grows faster and inverse leverage falls.

4 The main mechanism: dilution risk

The distinctive ingredient relative to fixed-yield rollover models is **dilution risk**. When fundamentals deteriorate, new creditors require a higher yield, i.e. higher R_t. But higher R_t implies faster growth in D_t (Eq. (3)), which dilutes existing creditors and accelerates the fall in x_t (Eq. (5)). This self-reinforcing loop makes runs more likely unless rollover rates are constrained by a cap.

5 Full value-function derivation (Appendix A)

This section rewrites the appendix derivation in one continuous chain, because it is the cleanest way to see how the run threshold is pinned down.

5.1 A.1 Timeline of events and three payoff cases

Fix a date $\tau \geq t$. A creditor who last lent at time $s \leq \tau$ holds debt with face value R_s. Let:

- τ_ϕ = conduit maturity time (asset pays out), Poisson intensity ϕ;
- τ_δ = debt-maturity time for the creditor, Poisson intensity δ;
- τ_θ = credit-line failure time *after* a run begins, Poisson intensity $\theta\delta$.

At any date τ, there are three mutually exclusive payoff events:

Case 1: conduit matures at $\tau = \tau_\phi$. The creditor is paid in full if assets cover debt, and otherwise receives a pro-rata share:

$$\frac{R_s}{D_{\tau_\phi}} \min\{D_{\tau_\phi}, y_{\tau_\phi}\} = R_s \min\left(1, \frac{y_{\tau_\phi}}{D_{\tau_\phi}}\right). \tag{12}$$

Case 2: conduit defaults at $\tau = \tau_\theta$. After a run starts, if credit lines fail the conduit liquidates and the creditor receives a pro-rata share of post-liquidation value:

$$\frac{R_s}{D_{\tau_\theta}} \min\{D_{\tau_\theta}, \ell_{y_{\tau_\theta}}\} = R_s \min\left(1, \frac{\ell_{y_{\tau_\theta}}}{D_{\tau_\theta}}\right), \tag{13}$$

where liquidation value (in present value units) is

$$\ell_{y_{\tau_\theta}} \equiv \alpha \frac{\phi}{\rho + \phi - \mu} y_{\tau_\theta}. \tag{14}$$

Case 3: the debt contract matures at $\tau = \tau_\delta$. At debt maturity the creditor chooses between *running* (take \$1 now) or *rolling over* into a new loan. Let $V(y_\tau, D_\tau, R_s; y^*)$ be the time-τ value of one dollar loaned at time $s \leq \tau$ given a conjectured run threshold y^*. Then the payoff at $\tau = \tau_\delta$ is

$$\max_{\text{roll over or run}} \{R_s V(y_{\tau_\delta}, D_{\tau_\delta}, R_{\tau_\delta}; y^*),\ R_s\} = R_s \max\{V(y_{\tau_\delta}, D_{\tau_\delta}, R_{\tau_\delta}; y^*),\ 1\}. \tag{15}$$

5.2 A.2 Creditor value function and homogeneity

Combining the three payoff cases, the time-t value to a creditor who last lent one dollar at time $s \leq t$ is

$$\begin{aligned} V(y_t, D_t, R_s; y^*) = \mathbb{E}_t&\left[e^{-\rho(\tau - t)} R_s \min\left(1, \frac{y_{\tau_\phi}}{D_{\tau_\phi}}\right) \mathbb{1}_{\{\tau = \tau_\phi\}}\right] \\ &+ \mathbb{E}_t\left[e^{-\rho(\tau - t)} R_s \min\left(1, \frac{\ell_{y_{\tau_\theta}}}{D_{\tau_\theta}}\right) \mathbb{1}_{\{\tau = \tau_\theta\}}\right] \\ &+ \mathbb{E}_t\left[e^{-\rho(\tau - t)} \max_{\text{roll over or run}} \{R_t V(y_{\tau_\delta}, D_{\tau_\delta}, R_{\tau_\delta}; y^*), R_s\} \mathbb{1}_{\{\tau = \tau_\delta\}}\right]. \end{aligned} \tag{16}$$

Now define inverse leverage $x_t \equiv y_t / D_t$. By homogeneity, Eq. (16) simplifies to

$$V(y_t, D_t, R_s; y^*) = R_s W(x_t; x^*), \tag{17}$$

where $x^* \equiv y^*/D^*$ is the *inverse-leverage* run threshold, and

$$\begin{aligned} W(x_t; x^*) = \mathbb{E}_t&\left[e^{-\rho(\tau - t)} \min(1, x_\tau) \mathbb{1}_{\{\tau = \tau_\phi\}}\right] \\ &+ \mathbb{E}_t\left[e^{-\rho(\tau - t)} \min(1, \ell x_\tau) \mathbb{1}_{\{\tau = \tau_\theta\}}\right] \\ &+ \mathbb{E}_t\left[e^{-\rho(\tau - t)} \max_{\text{roll over or run}} \{R_t W(x_\tau; x^*), 1\} \mathbb{1}_{\{\tau = \tau_\delta\}}\right], \end{aligned} \tag{18}$$

and $\ell \equiv \alpha \frac{\phi}{\rho + \phi - \mu}$ is the liquidation scaling in inverse-leverage units.

Finally, applying Itô and Eq. (3) yields the inverse-leverage dynamics (reproduced in the appendix):

$$\frac{\mathrm{d}x_t}{x_t} = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}Z_t + \delta \, \mathrm{d}t - \delta R_t \, \mathrm{d}t. \tag{19}$$

5.3 A.3 Competitive break-even, yield cap, and the run threshold

Break-even. In a competitive market, for each \$1 invested at t creditors break even:

$$1 = R_t W(x_t; x^*). \tag{20}$$

Cap. Because face values cannot exceed $\bar{R}$, equilibrium rollover promises satisfy

$$R_t = \min\Big\{\bar{R},\ W(x_t; x^*)^{-1}\Big\}. \tag{21}$$

Run threshold. Runs occur when the needed break-even promise would exceed the cap. Equivalently, the threshold x^* is pinned down by the boundary condition

$$\bar{R} = W(x^*; x^*)^{-1}. \tag{22}$$

5.4 A.4 Proposition 1 and proof (the threshold logic)

Proposition 1. Let $R_t = \min\{\bar{R}, W(x_t; x^*)^{-1}\}$. Then

$$R_t = \begin{cases} W(x_t; x^*)^{-1}, & x_t > x^*, \\ \bar{R} = W(x^*; x^*)^{-1}, & x_t = x^*, \\ \bar{R}, & x_t < x^*. \end{cases} \tag{23}$$

Proof. At debt maturity, the creditor's continuation payoff (per \$1 face value) is

$$\begin{aligned} \max_{\text{run or roll}} \{1,\ R_t W(x_t; x^*)\} &= \max_{\text{run or roll}} \Big\{1,\ \min\big(\bar{R}, W(x_t; x^*)^{-1}\big)\, W(x_t; x^*)\Big\} \\ &= \max_{\text{run or roll}} \Big\{1,\ \min\big(\bar{R}\, W(x_t; x^*), 1\big)\Big\}. \end{aligned} \tag{24}$$

Step 1: show that if $x_t < x^*$, creditors run. If $x_t < x^*$, then by definition of x^*,

$$\min\big(\bar{R}, W(x_t; x^*)^{-1}\big)\, W(x_t; x^*) < 1. \tag{25}$$

So rolling over yields strictly less than 1, while running yields 1, hence creditors run.

Step 2: show that if $x_t \geq x^*$, creditors roll over. If $x_t \geq x^*$, creditors choose to roll over; otherwise their payoff would be 1. Since their payoff is bounded above by 1, rolling over must also pay 1:

$$\min\big(\bar{R}, W(x_t; x^*)^{-1}\big)\, W(x_t; x^*) = 1 \implies \min\big(\bar{R}\, W(x_t; x^*), 1\big) = 1. \tag{26}$$

Step 3: pin down the boundary. Because $W(\cdot; x^*)$ is strictly increasing in x, the set $\{x :$

$\bar{R}\,W(x;x^*) \geq 1\}$ is an interval. Thus either:

$$R_t = \begin{cases} W(x_t;x^*)^{-1} > \bar{R}, & x_t \geq x^*, \\ \bar{R}, & x_t < x^*, \end{cases} \qquad \text{(case i)} \tag{27}$$

$$\text{or} \quad R_t = \begin{cases} W(x_t;x^*)^{-1}, & x_t > x^*, \\ \bar{R}, & x_t = x^*, \\ \bar{R}, & x_t < x^*, \end{cases} \qquad \text{(case ii)}. \tag{28}$$

Case (i) cannot hold. In case (i) we would have $R^* \equiv W(x^*;x^*)^{-1} < \bar{R}$:

$$R^* = W(x^*;x^*)^{-1} < \bar{R}, \tag{29}$$

but at the boundary we must have exactly

$$1 = R^* W(x^*;x^*) < \bar{R}\,W(x^*;x^*), \tag{30}$$

which contradicts continuity and the definition of x^* as the run threshold. Therefore case (ii) holds, proving Eq. (23).

□

6 Analytical solution below the run threshold (Appendix A.4)

Given x^*, the value function solves an HJB that reflects three Poisson events (conduit maturity, debt maturity, credit-line failure) and the Brownian evolution in between.

6.1 HJB equation

Using Eqs. (18) and (19), the HJB is

$$\begin{aligned} \rho W(x;x^*) = {} & \big[\mu - \delta(R_t - 1)\big] x W_x(x;x^*) + \frac{\sigma^2}{2} x^2 W_{xx}(x;x^*) \\ & + \phi\big[\min(1,x) - W(x;x^*)\big] + \theta\delta\, \mathbb{1}_{\{x \leq 1/\ell\}} \big[\min(1,\ell x) - W(x;x^*)\big] \\ & + \delta\Big[\max\{R_t W(x;x^*), 1\} - W(x;x^*)\Big]. \end{aligned} \tag{31}$$

6.2 General solution form for $x < x^*$

Below the threshold, the cap binds so $R_t = \bar{R}$. The general solution to the ODE implied by (31) is

$$W(x;x^*) = d_1 x^{\eta} + d_2 x^{-\gamma} - \frac{a_5}{a_3} - \frac{a_4}{a_3 + a_1} x, \qquad x \leq x^*, \tag{32}$$

where

$$\eta = \frac{1}{2a_2}\left(a_2 - a_1 + \sqrt{(a_2-a_1)^2 - 4a_3a_2}\right) > 1,$$
$$-\gamma = \frac{1}{2a_2}\left(a_2 - a_1 - \sqrt{(a_2-a_1)^2 - 4a_3a_2}\right) < -1,$$

and coefficients are

$$a_1 = \mu + \delta - \delta\bar{R}, \qquad a_2 = \frac{\sigma^2}{2} > 0, \qquad a_3 = -(\phi + \rho + \theta\delta + \delta) < 0,$$
$$a_4 = \alpha\theta\delta\, \mathbb{1}_{\{x \le 1/\ell\}} + \phi\, \mathbb{1}_{\{x \le 1\}} \ge 0, \qquad a_5 = \delta + \theta\delta\, \mathbb{1}_{\{x > 1/\ell\}} + \phi\, \mathbb{1}_{\{x > 1\}} > 0.$$

6.3 Boundary conditions

The constants (d_1, d_2) are pinned down by value matching and smooth pasting at $x = 1$, $x = 1/\ell$, and at x^*, plus limits:

$$W_x(0; x^*) = \frac{\theta\delta\left(\frac{\alpha\phi}{\phi+\rho-\mu}\right) + \phi}{\delta\bar{R}(\phi + \rho - \mu) + \theta\delta} > 0, \tag{33}$$
$$W(0; x^*) = \frac{\delta}{\phi + \rho + \theta\delta} > 0, \tag{34}$$
$$\lim_{x\to\infty} W(x; x^*) = \frac{\phi + \delta}{\phi + \delta + \rho}. \tag{35}$$

6.4 Closed forms in two main cases

The paper reports explicit expressions depending on where x^* lies.

Case 1: $0 < x^* \le 1$**.** Then

$$W(x; x^*) = A_1 x^\eta - \frac{a_5}{a_3} - \frac{a_4}{a_3 + a_1} x \qquad (x \le x^*), \tag{36}$$

with

$$A_1 = \left(\frac{1}{\bar{R}} + \frac{a_5}{a_3}\right)(x^*)^{-\eta} + \frac{a_4}{a_3 + a_1}(x^*)^{1-\eta}.$$

Case 2: $1 < x^* \le 1/\ell$**.** Then

$$W(x; x^*) = \begin{cases} A_2 x^\eta - \frac{a_5}{a_3} + \frac{a_4}{a_3+a_1} x, & x \le 1, \\ B_1 x^\eta + B_2 x^{-\gamma} - \frac{b_5}{a_3} - \frac{b_4}{a_3+a_1} x, & 1 < x \le x^*, \end{cases} \tag{37}$$

where $b_4 = \phi$ and $b_5 = \delta + \theta\delta$, and (A_2, B_1, B_2) are determined by matching/smooth-pasting at $x = 1$ and $x = x^*$ (the explicit expressions are given in the paper).

Case 3: $x^* > 1/\ell$**.** This case would imply that creditors run even when liquidation would still repay them in full, which the paper argues is economically implausible in their setting; the quantitative results focus on cases 1–2.

7 Numerical algorithm (Appendix B)

To solve for the equilibrium threshold x^*, the algorithm is:

1. Guess a value of x^*.
2. Solve analytically for $W(x; x^*)$ for $x \le x^*$ using Appendix A.
3. Solve numerically for $W(x; x^*)$ for $x > x^*$.
4. Check the boundary condition $W(x^*; x^*) = 1/\bar{R}$; if not satisfied, update the guess for x^*.

For step 3, reduce the ODE order by defining $Z \equiv W_x$:

$$W_x = Z, \qquad Z_x = W_{xx}, \tag{38}$$

and then use the HJB to write Z_x as a function of (W, Z, x):

$$\begin{aligned} Z_x = &-\frac{2(\mu+\delta)}{\sigma^2 x} Z + \frac{2\delta}{\sigma^2 x} Z - \frac{2\phi}{\sigma^2 x^2} \min(1, x) \\ &+ \frac{2(\rho+\phi+\delta)}{\sigma^2 x^2} W - \frac{2\delta}{\sigma^2 x^2}. \end{aligned} \tag{39}$$

(Here $R_t = W(x; x^*)^{-1}$ for $x > x^*$.)

8 Empirical and estimation "models" in the main text

In addition to the structural model, the paper defines several regression-based objects that become SMM moments and also provide diagnostic figures.

8.1 Yield-volatility regression (moments M3–M4)

$$|r_{i,t} - r_{i,t-1}| = \beta_0 + \beta_1 r_{i,t-1} + \varepsilon_{i,t}. \tag{6}$$

Model-implied (instantaneous) yield volatility is

$$\mathrm{var}_t(\mathrm{d}r_t) = \left(x_t \frac{\partial r}{\partial x}(x_t; x^*) \right)^2 \sigma^2 \, \mathrm{d}t. \tag{7}$$

8.2 Event-time yield dynamics before runs (moments M5–M7)

Let $\tau_{i,t}$ be "weeks relative to run" (negative values are pre-run).

$$r_{i,t} = \gamma_0 + \gamma_1 \tau_{i,t} + \gamma_2 e^{\tau_{i,t}} + \varepsilon_{i,t}. \tag{8}$$

8.3 Run forecasting regressions (moments M8–M13)

For horizons $\tau \in \{2, 4, 8\}$ weeks:

$$\mathbb{1}\{\text{run within } \tau \text{ weeks of } r_{i,t}\} = \lambda_0^\tau + \lambda_1^\tau \frac{r_{i,t}}{\max r_i} + \varepsilon_{i,t}. \tag{9}$$

Here $\max r_i$ proxies for the conduit-specific cap $\bar{r}_i$ (the maximum yield spread the conduit ever reaches in the sample).

8.4 Debt-quantity accounting (used for Fig. 7)

The paper decomposes total face value D_{it} into market-held ABCP D_{it}^M and sponsor-held debt D_{it}^S, so $D = D^M + D^S$. Let $\mathbb{1}_{\{run_{it}\}}$ be the run indicator in week t.

Using (3), the components evolve as

$$\frac{\mathrm{d}D_{it}^M}{\mathrm{d}t} = -\delta D_{it}^M + (1 - \mathbb{1}_{\{run_{it}\}})\delta D_{it} R_{it}, \tag{10}$$

$$\frac{\mathrm{d}D_{it}^S}{\mathrm{d}t} = -\delta D_{it}^S + \mathbb{1}_{\{run_{it}\}}\delta D_{it} \bar{R}_i. \tag{11}$$

Interpretation:

- When not in a run, maturing debt is replaced by new market ABCP issuance (face value R_{it}).
- When in a run, market issuance stops and the sponsor effectively funds maturities at the cap $\bar{R}_i$.

9 SMM estimator and asymptotics (Appendix C)

Denote empirical moments by $h(x_i)$ and simulated moments by $h(y_{i,k}(b))$ for simulation draw $k = 1, \ldots, K$ and parameter vector b. The SMM estimator is

$$\hat{b} = \arg\min_b \; g_n(b)^\top \widehat{W}_n g_n(b), \tag{40}$$

where

$$g_n(b) = \frac{1}{n} \sum_{i=1}^n \left[h(x_i) - \frac{1}{K} \sum_{k=1}^K h(y_{i,k}(b)) \right]. \tag{41}$$

Under standard regularity conditions,

$$\sqrt{n}(\hat{b} - b) \xrightarrow{d} \mathcal{N}(0,\ \text{var}(b)), \tag{42}$$

with

$$\text{var}(b) = \left(1 + \frac{1}{K}\right) \left[\frac{\partial g_n(b)'}{\partial b} \widehat{W} \frac{\partial g_n(b)}{\partial b} \right]^{-1} \left[\frac{\partial g_n(b)'}{\partial b} \widehat{W} \, \Omega \, \widehat{W} \frac{\partial g_n(b)}{\partial b} \right] \left[\frac{\partial g_n(b)'}{\partial b} \widehat{W} \frac{\partial g_n(b)}{\partial b} \right]^{-1}. \tag{43}$$

The paper estimates Ω using a GMM/sandwich approach that allows heteroskedasticity and serial correlation across observations close in calendar time.

10 Calibration choices (what is fixed before estimation)

Key fixed inputs (baseline choices in the paper):

- ρ: set to 4.9% (annualized 1-month T-bill rate at start of 2007).
- δ: set from mean ABCP maturity (37 days), i.e. $1/\delta = 0.101$ years.
- ϕ: set from average conduit age (5.8 years), i.e. $\phi = 1/5.8$.
- μ: not identified; baseline sets $\mu = \rho$ (robustness checks use $\mu = \rho \pm 1\%$).

They do not estimate initial inverse leverage x_0; instead they condition moment calculation on observations with spreads above 10 b.p. to obtain informative data where spreads respond to leverage.

11 Interpreting all tables and figures

11.1 Figure 1: fundamentals deteriorate and runs rise

Figure 1 overlays weekly price indices for key asset categories and the fraction of programs in a run week. Two facts stand out:

1. Some asset categories decline during 2007, consistent with worsening fundamentals.
2. Runs cluster in late 2007: the fraction of programs in a run week rises sharply after August.

This motivates a fundamentals-driven run model where yields increase before runs.

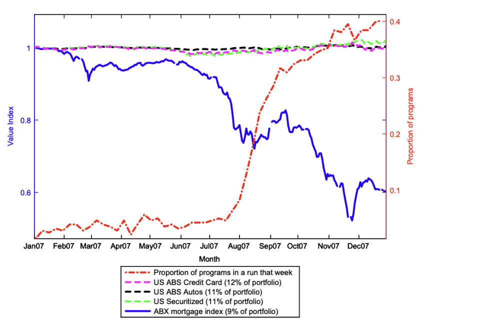

Fig. 1. This figure shows the weekly series of prices for several asset categories in the portfolio of ABCP conduits in 2007 (dashed and solid lines), as well as the proportion of ABCP programs experiencing runs in a given week (dashed-dot line). We normalize prices to $1 on January 1, 2007. The ABX index of mortgage securities is an average across tranches (from BBB- to AAA) of the ABX.HE indexes for MBS originated in the first half of 2006. Data for the remaining asset categories are from Barclays indexes. 'US Securitized' is an aggregate of U.S. asset-backed securities, commercial mortgage-backed securities, and other mortgage-backed securities; this index proxies for the ambiguous 'Securities' category, which makes up 11% of conduit assets in 2007. Portfolio weights in the legend are from 'The ABC's of ABCP,' an unpublished document from Societe Generale. Data for the proportion of runs are from the DTCC database on all issues by ABCP programs, where a run is defined as in Covitz, Liang, and Suarez (2013): an ABCP program experiences a run in a given week if either (1) more than 10% of the program's outstanding paper is scheduled to mature, yet the program does not issue new paper; or (2) the program was in a run the previous week and it does not issue new paper in the current week.

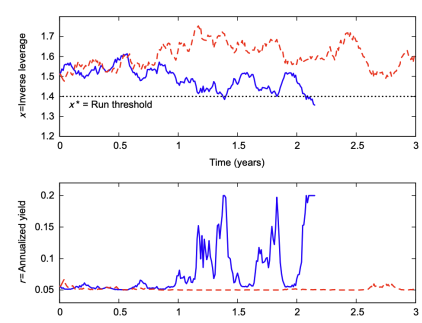

Fig. 2. This figure shows the simulated paths of two conduits with the same initial leverage and parameter values. The top panel shows simulated values of x_t, inverse leverage. The dotted line denotes the run threshold. The bottom panel shows simulated paths of annual yields at rollover for the same two conduits. The risk-free rate is 5% and the cap on the rollover yield is 20%.

11.2 Figure 2: simulated paths (mechanism visualization)

Figure 2 simulates two conduits: one path stays above x^* (no run, low and stable spreads), the other drifts toward x^* (spreads rise and become volatile), and then runs when the cap binds. This figure makes Eq. (5) concrete: higher spreads imply higher R_t, which lowers x_t faster.

11.3 Table 1: flexible yields vs. fixed-yield He–Xiong

Table 1 compares one-year run probabilities and thresholds between this model (flexible yields) and He–Xiong (fixed yields). Panel A reports the ratio of one-year run probability in this model to that in He–Xiong.

What we take from the numbers.

- Under He–Xiong calibrated parameters, the ratios are often **well above 1** (e.g. for capped rollover yields of 15–25% and fixed yields of 7–9%, the run-probability ratios are about 3.6 and can exceed 40–50).
- The threshold ratio in Panel B is also > 1 (often around 1.4–2.3), meaning the run threshold is *higher* (runs occur at safer states) when yields are flexible.
- Under the paper's estimated parameters, the ratio can be below 1 for very low fixed yields (e.g. about 0.65 for the lowest fixed-yield case), because low early borrowing costs can delay fragility.

Overall, Table 1 is the quantitative evidence that **dilution risk is a first-order amplification channel**.

Table 1
The effect of flexible yields on runs.

This table compares the predictions from our model to the predictions of He and Xiong (2012a), denoted HX. Yields change over time in our model, whereas yields are constant in HX. The columns on the left use HX's calibrated values: $\rho = 1.5\%$, $\phi = 0.077$, $\alpha = 55\%$, $\sigma = 20\%$, $\mu = 1.5\%$, $y_0 = 1.4$, $\delta = 10$, and $\theta = 5$. The columns on the right use estimated parameter values for the weak-guarantee subsample in Table 4. Panel A shows the fraction of simulated conduits that experience a run in our model within one year, divided by the corresponding fraction from HX. Panel B shows the run threshold in our model (x^*) divided by the run threshold in HX. Panel C shows the conduit's initial inverse leverage in our model (x_0) divided by the conduit's initial inverse leverage in HX; these results are identical out to two digits for the three values of $\bar{r}$. The parameter $\bar{r}$ is our model's cap on yield spreads, and ρ is the risk-free rate, so $\bar{r}+\rho$ is the capped rollover yield.

	Using HX parameters			Using estimated parameters		
Panel A: Ratio of the one-year run probability in our model to that in HX						
HX's fixed yield:	5%	7%	9%	$\rho+0.1\%$	$\rho+0.3\%$	$\rho+0.5\%$
$\bar{r}+\rho=15\%$	1.31	3.73	50.64	0.65	1.48	3.61
$\bar{r}+\rho=20\%$	1.29	3.64	49.08			
$\bar{r}+\rho=25\%$	1.28	3.56	47.65			
Panel B: Ratio of the run threshold in our model to that in HX						
HX's fixed yield:	5%	7%	9%	$\rho+0.1\%$	$\rho+0.3\%$	$\rho+0.5\%$
$\bar{r}+\rho=15\%$	1.43	1.77	2.29	1.22	1.24	1.26
$\bar{r}+\rho=20\%$	1.42	1.75	2.28			
$\bar{r}+\rho=25\%$	1.41	1.74	2.26			
Panel C: Ratio of initial inverse leverage in our model to that in HX						
HX's fixed yield:	5%	7%	9%	$\rho+0.1\%$	$\rho+0.3\%$	$\rho+0.5\%$
	1.38	1.54	1.70	1.25	1.26	1.27

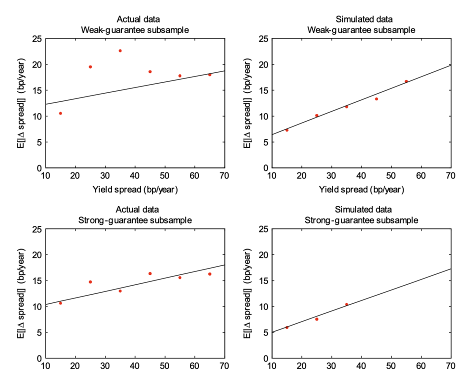

Fig. 3. This figure shows the relation between yield spread volatility and the yield spread level, both measured in basis points per year (bp/year). The vertical axis is the absolute value of one-week changes in yield spread. The horizontal axis is the lagged yield spread. The left-hand (right-hand) panels show results in actual (simulated) data. The top panels show results for the weak-guarantee subsample, which contains the 90 ABCP conduits in 2007 with extendible or SIV guarantees. The bottom panels show results for the strong-guarantee subsample, which contains the 191 ABCP conduits in 2007 with full credit or full liquidity credit guarantees. The points show local averages, and the solid line shows predicted values from a regression of absolute changes in yield spreads on the lagged yield spread. The intercept and slope from this regression provide two of the 13 moments used in the SMM estimation. The reason there are fewer points in the bottom-right panel compared to the bottom-left panel is that the estimated cap on yield spread is 36 basis points in the strong-guarantee subsample; all simulated spreads are therefore ≤ 36 basis points, whereas there are a few spreads > 36 basis points in the actual data.

11.4 Figure 3: volatility of spreads increases in spread level

Figure 3 plots the relation between $|r_t - r_{t-1}|$ and r_{t-1}, in data and in the simulated model, for weak and strong guarantee subsamples. The increasing pattern aligns with Eq. (7): near the cap, $|\partial r/\partial x|$ is large, so small shocks to x_t translate into large spread moves.

11.5 Figure 4: spreads rise nonlinearly in the weeks before runs

Figure 4 plots average spreads in event time (weeks to run start), along with the fitted curve from Eq. (8). The convex pattern is important for identifying $\bar{r}$ (the cap shapes how fast spreads can rise as the state deteriorates).

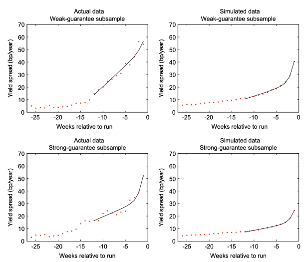

Fig. 4. This figure plots average yield spreads (basis points per year, bp/year) in event time leading up to runs in event week zero. The left-hand (right-hand) panels show results in actual (simulated) data. The top panels show results for the weak-guarantee subsample, which contains the 90 ABCP conduits in 2007 with extendible or SIV guarantees. The bottom panels show results for the strong-guarantee subsample, which contains the 191 ABCP conduits in 2007 with full credit or full liquidity credit guarantees. The points show the average yield spread in each week. The solid line shows the predicted values from the regression of yield spreads on the number of weeks relative to the run and the exponent of that same number. The intercept and two slopes from this regression provide three of the 13 moments used in SMM estimation. The solid line starts at week -12 because the regression only uses data from weeks -12 to -1.

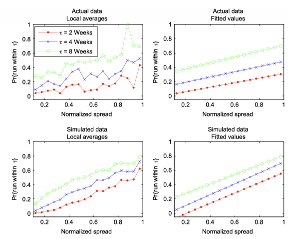

Fig. 5. This figure shows the forecasting relation between yield spreads and runs in the weak-guarantee subsample, which contains 90 ABCP conduits in 2007 with extendible or SIV credit guarantees. The top panels show the relationship using actual data, and the bottom panels show the relationship using data simulated from the model with the parameter estimates in Table 4. The vertical axis is the probability of a run within τ weeks of the current conduit/week observation. The horizontal axis is the normalized yield, defined as the current yield spread divided by *maxr*, a proxy for the conduit's cap on yield spreads. The left-hand panels show local averages of run probabilities. The right-hand panels show the predicted values from a regression of $\mathbf{1}_{\{\text{run within } \tau \text{ weeks}\}}$ on the normalized yield spread. We estimate this regression for forecasting horizons of $\tau = 2$, 4, and 8 weeks. The intercepts and slopes from these regressions provide six of the 13 moments used in SMM estimation.

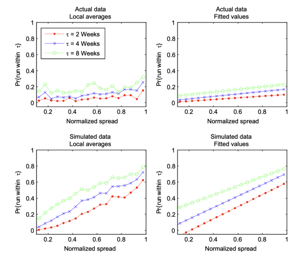

Fig. 6. This figure shows the forecasting relation between yield spreads and runs in the strong-guarantee subsample, which contains the 191 ABCP conduits in 2007 with full credit or full liquidity guarantees. Definitions are the same as in Fig. 5.

11.6 Figures 5–6: spreads forecast runs (and model fit differs by subsample)

Figures 5 and 6 show the probability of a run within 2, 4, or 8 weeks as a function of normalized spreads. These figures visualize the moments in Eq. (9).

Reading:

- The **weak-guarantee subsample** exhibits a strong and fairly smooth forecasting relation that the model fits reasonably well.
- The **strong-guarantee subsample** has fewer runs and more heterogeneity (e.g. different caps and rarely binding constraints), so a representative-parameter model tends to overpredict how strongly high spreads translate into run probabilities.

11.7 Table 2: Jacobian elasticities (identification map)

Table 2 reports elasticities of each moment with respect to each parameter $(\theta, \sigma, \bar{r}, \alpha)$ (evaluated at the weak-guarantee estimate). The table is a compact identification story:

- θ mainly moves recovery probability (Moment 1): guarantees matter *after* runs start.
- σ strongly moves expected recovery time (Moment 2): volatility changes how quickly the state returns above the threshold.
- $\bar{r}$ is crucial for the pre-run yield-shape moments (Moments 5–7): it governs how spreads evolve as the cap approaches.
- α strongly moves the run-forecasting moments (Moments 8–11): liquidation losses change the run-vs-roll tradeoff.

The paper reports full rank and a low condition number for the Jacobian in both subsamples, supporting local identification.

Table 2
Estimated Jacobian matrix.

This table presents the estimates of the Jacobian matrix for the 13 moment conditions in our SMM estimation procedure, for the subsample of 90 ABCP conduits in 2007 with SIV or extendible credit guarantees. Moment 1 is the probability that a conduit experiences a recovery within eight weeks of a run's start. Moment 2 is the average number of days between the run's start and recovery, conditional on a recovery occurring within eight weeks of the run's start. Moments 3 and 4 are the intercept and slope from a regression of absolute changes in yield spreads on the lagged yield spread. Moments 5–7 are the intercept and slopes from a regression of yield spreads on the number of weeks relative to a run and the exponent of that same number. Moments 8–13 come from three regressions, each of the indicator $\mathbf{1}_{\{\text{run within } \tau \text{ weeks}\}}$ on the current yield spread. The three regressions use $\tau=2$, 4, and 8 weeks. Each row of each matrix contains the estimated elasticities of the given moment with respect to the parameters across its columns. Parameters are estimated by SMM, which chooses values that minimize the distance between actual and simulated moments. Section 2 describes the model used to simulate moments. The number highlighted in each row in bold type face corresponds to the moment's highest elasticity in absolute value.

		Elasticity of moments with respect to			
		θ	σ	$\bar{r}$	α
Moments on time between run and recovery (τ):					
1.	$\Pr[\tau < 8$ weeks$]$	**−0.209**	−0.106	−0.036	0.075
2.	$E[\tau \mid \tau \le 8$ weeks$]$ (in days)	−0.104	**−0.295**	−0.011	0.106
Moments from regression of $\lvert r_{it+1} - r_{it} \rvert$ on r_{it}:					
3.	Intercept	0.112	0.106	0.724	**−0.758**
4.	Slope	0.002	−0.307	0.133	**0.472**
Moments describing yield spreads leading up to runs:					
Regression of r_{it} on $\tau[\equiv$ weeks relative to run$]$ and $\exp(\tau)$					
5.	Intercept	0.102	−0.186	**0.990**	−0.311
6.	Slope on τ	0.133	−0.149	**0.977**	−0.870
7.	Slope on $\exp(\tau)$	0.235	−0.548	**1.093**	−0.118
Regressions of $\mathbf{1}_{\{\text{run within } \tau \text{ weeks}\}}$ on yield spread:					
8.	Intercept ($\tau=2$)	0.023	−0.424	0.243	**−0.446**
9.	Slope ($\tau=2$)	−0.030	0.177	−0.055	**−0.296**
10.	Intercept ($\tau=4$)	−0.414	0.694	−0.498	**−2.562**
11.	Slope ($\tau=4$)	0.028	−0.023	0.057	**−0.444**
12.	Intercept ($\tau=8$)	−0.158	0.283	**−0.571**	−0.344
13.	Slope ($\tau=8$)	0.066	−0.067	**0.180**	−0.097

Table 3
Moments from SMM estimation.

This table shows the 13 moments used in SMM estimation. The first (last) four columns show moments for the weak- (strong-) guarantee subsample, which contains 90 ABCP conduits with extendible or SIV credit guarantees (191 conduits with full credit or full liquidity guarantees). Simulated moments are computed using the parameter estimates in Table 4. Moment 1 is the probability that a conduit recovers within eight weeks of a run's start. Moment 2 is the average number of days until the recovery, provided it occurs within eight weeks of the run's start. Moments 3 and 4 are the intercept and slope from a regression of absolute changes in yield spreads on the lagged yield spread. Moments 5–7 are the intercept and slopes from a regression of yield spreads on the number of weeks relative to a run and its exponential. Moments 8–13 come from three regressions, each of the indicator $\mathbf{1}_{\{\text{run within } \tau \text{ weeks}\}}$ on the current yield spread, where $\tau=2$, 4, and 8 weeks. The moment condition *t*-statistic tests whether the actual and simulated moments are equal. The *J*-test is the χ^2 test for the model's overidentifying restrictions.

	Weak-guarantee subsample				Strong-guarantee subsample			
	Actual moments		Simulated moments estimate	Moments condition *t*-stat.	Actual moments		Simulated moments estimate	Moments condition *t*-stat.
	Estimate	Std. Err.			Estimate	Std. err.		
Moments on time between run and recovery (τ):								
1. $\Pr[\tau < 8$ weeks$]$	0.451	0.054	0.513	−1.12	0.613	0.060	0.615	−0.04
2. $E[\tau \mid \tau \le 8$ weeks$]$ (in days)	17.1	1.4	15.4	1.16	17.4	0.7	17.4	0.04
Moments from regression of $\lvert r_{it+1} - r_{it} \rvert$ on r_{it}:								
3. Intercept	0.0011	0.0006	0.0004	1.21	0.0009	0.0004	0.0003	1.58
4. Slope	0.108	0.078	0.223	−1.48	0.128	0.060	0.205	−1.29
Moments describing yield spreads leading up to runs:								
Regression of r_{it} on $\tau[\equiv$ weeks relative to run$]$ and $\exp(\tau)$								
5. Intercept	0.00538	0.00190	0.00248	1.52	0.00347	0.00130	0.00157	1.47
6. Slope on τ	0.00033	0.00012	0.00012	1.80	0.00015	0.00010	0.00007	0.85
7. Slope on $\exp(\tau)$	0.00168	0.00268	0.00467	−1.12	0.00525	0.00178	0.00260	1.49
Moments from regressions of $\mathbf{1}_{\{\text{run within } \tau \text{ weeks}\}}$ on yield spreads								
8. Intercept ($\tau=2$)	−0.003	0.036	−0.146	3.94	−0.001	0.012	−0.158	9.64
9. Slope ($\tau=2$)	0.317	0.087	0.716	−4.54	0.104	0.040	0.755	−14.93
10. Intercept ($\tau=4$)	0.121	0.055	−0.044	2.97	0.016	0.013	0.000	0.93
11. Slope ($\tau=4$)	0.364	0.045	0.758	−8.46	0.155	0.041	0.711	−12.37
12. Intercept ($\tau=8$)	0.297	0.095	0.149	1.55	0.067	0.028	0.217	−4.89
13. Slope ($\tau=8$)	0.413	0.073	0.655	−3.24	0.165	0.034	0.561	−10.02
Test of over-identifying restrictions:								
J-test	1,111				6,771			
p-Value	0.000				0.000			

11.8 Table 3: fit of the 13 moments

Table 3 compares empirical moments to simulated moments at $\hat{b}$.

What fits well. Recovery moments (M1–M2) are close in both subsamples; the model also reproduces the qualitative rise in yield volatility and in pre-run spreads.

Where the model is "too sharp." In the forecasting regressions (M8–M13), simulated slopes are often substantially larger than empirical slopes (especially in the strong-guarantee subsample). This is consistent with the idea that real conduits differ in caps/guarantee details and the model uses a single representative parameter vector.

11.9 Figure 7: aggregate ABCP outstanding (out-of-sample check)

Figure 7 compares actual total ABCP outstanding in 2007 to the model's predicted total debt D_t (market ABCP plus sponsor debt). A key mechanism is that runs shift funding from market ABCP to sponsor balance sheets, captured by Eqs. (10)–(11). The model explains a large fraction of the contraction in total ABCP after August 2007 (reported as 73%).

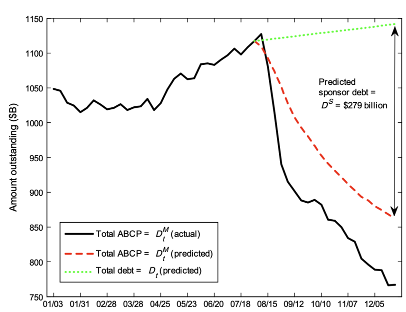

Fig. 7. This figure shows the total face value of debt outstanding during 2007. The solid line shows actual data for D_t^M, the amount of ABCP outstanding in our sample of 281 conduits. The dashed red line shows predicted values for D_t^M. The dotted green line shows the predicted value of total debt D_t, which is the sum of market ABCP debt, D^M, and debt held by the sponsor, D^S. The figure also notes that the predicted total face value of debt held by sponsors at the end of 2007 is $279 billion. We predict debt quantities after August 1, 2007 using Eqs. (10) and (11) with estimated parameters from Table 4, conduits' total face value of ABCP on August 1, and conduit/week data on rollover yields and run status throughout 2007. We assume sponsors held zero debt on August 1.

Table 4
Structural parameter estimates and predicted run thresholds.

The first two rows provide parameter estimates in the weak-guarantee subsample, which contains 90 ABCP conduits in 2007 with SIV or extendible credit guarantees. The last two rows report estimates in the strong-guarantee subsample, which contains 191 ABCP conduits in 2007 with full credit or full liquidity guarantees. Columns 2–5 report estimated structural parameters, with standard errors in parentheses. Estimation is done by SMM, which chooses parameter estimates that minimize the distance between actual and simulated moments. Section 2 describes the model used to simulate moments. Standard errors account for time-series and cross-sectional autocorrelation, both within and across the 13 moments used in estimation. Standard errors also include a two-step correction for measurement errors in parameters δ (inverse average debt maturity) and ϕ (inverse asset maturity). The last column shows the leverage threshold for runs ($1/x^*$) predicted by the model for the given parameter estimates; the model predicts that runs occur as soon as leverage exceeds this threshold.

	Parameter estimates				Predicted
Subsample	Weakness of credit guarantee θ	Asset volatility (% per year) σ	Cap on yield spreads (b.p. per year) $\bar{r}$	Asset liquidity α	leverage threshold for runs (%) $1/x^*$
Weak guarantees	0.449 (0.133)	4.30 (0.10)	59.8 (6.7)	0.920 (0.032)	92.0
Strong guarantees	0.141 (0.045)	3.54 (0.07)	36.0 (7.0)	0.968 (0.045)	97.1

11.10 Table 4: parameter estimates and implied run thresholds

Table 4 reports estimates separately by subsample:

- **Guarantees:** θ is much higher under weak guarantees ($\hat{\theta} = 0.449$) than strong guarantees ($\hat{\theta} = 0.141$).
- **Volatility:** σ is higher for weak guarantees (4.30%) than for strong (3.54%).
- **Yield cap:** $\bar{r}$ is higher for weak guarantees (about 60 b.p. per year) than for strong (about 36 b.p.).
- **Liquidity:** α is high in both (0.920 vs 0.968), but small differences matter a lot for fragility.
- **Implied run thresholds:** leverage at the run threshold, $1/x^*$, is extremely high: 92% (weak) and 97.1% (strong). This means conduits are run-prone even while "nearly solvent."

11.11 Figure 8 and Table 5: fragility and sensitivity

Figure 8. Run probabilities are extremely sensitive to initial leverage and to liquidity α. The figure also shows sensitivity to parameters like maturity ($1/\delta$), volatility σ, and the yield cap $\bar{r}$, but leverage/liquidity dominate.

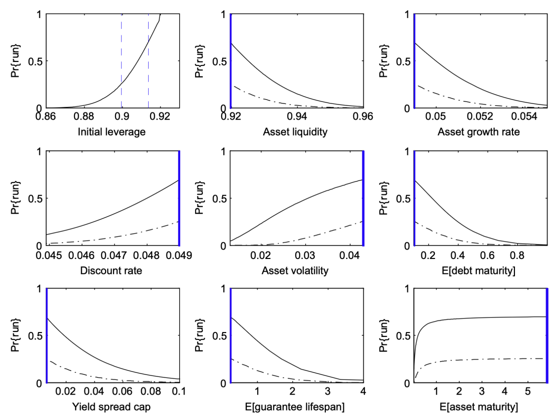

Fig. 8. This figure shows the sensitivity of run probabilities to the model's parameters: initial conduit leverage ($1/x_0$), asset liquidity (α), asset growth rat (μ), discount rate (ρ), asset volatility (σ), expected debt maturity ($1/\delta$), yield spread cap ($\bar{r}$), expected guarantee lifespan ($1/(\delta\theta)$), and expected asse maturity ($1/\phi$). Expected guarantee lifespan and debt maturity are in years. Asset growth rate and volatility, the discount rate, and the cap on yield spread are in units of fraction per year. Each panel plots the simulated three-month run probability as a function of parameter values. In each panel we se parameter values to their estimates for the weak-guarantee subsample (Table 4), then we vary the parameter indicated on the panel's horizontal axis. Initia leverage (top left panel) serves as each simulation's initial condition. The two vertical dashed blue lines in the top-left panel indicate the values we use a initial conditions in the remaining panels. The bold blue lines on the vertical axes indicate the parameters' estimated value. The solid (dashed-dot) lin shows run probabilities when initial leverage is at the high (low) level indicated in the top-left panel.

Table 5
The change in asset value required to trigger a run.

This table shows simulated changes in asset value preceding runs. We simulate conduits for one year using parameter estimates for the weak-guarantee subsample (Table 4). For these parameter values, runs occur as soon as leverage exceeds 0.92. Panel A contains the simulations' assumed initial leverage, which equals the asset's fundamental value y_0 divided by the initial face value of debt D_0. Panel B shows statistics on the percentage change in asset value y_t between the simulation's start and the run's start, for those conduits that experience a run.

Panel A: Initial leverage (D_0/y_0)							
	0.85	0.86	0.87	0.88	0.89	0.9	0.91
Panel B: Percent change in asset value preceding a run							
Mean	−4.39	−3.53	−2.69	−1.93	−1.26	−0.68	−0.26
25th percentile	−4.97	−4.22	−3.53	−2.85	−2.24	−1.62	−0.95
Median	−4.28	−3.47	−2.70	−2.04	−1.50	−1.05	−0.67
75th percentile	−3.70	−2.76	−1.85	−1.04	−0.42	0.00	0.02

Table 5 (how big a shock triggers a run). Table 5 translates sensitivity into a fragility statistic: the required asset-value drop to trigger a run. When initial leverage is high ($D_0/y_0 \approx 0.90$), a mean decline of under 1% can be sufficient; at lower leverage (0.85), the required decline is closer to 4–5%. So the model implies a **nonlinear fragility**: small increases in leverage can drastically reduce the shock size needed to produce a run.

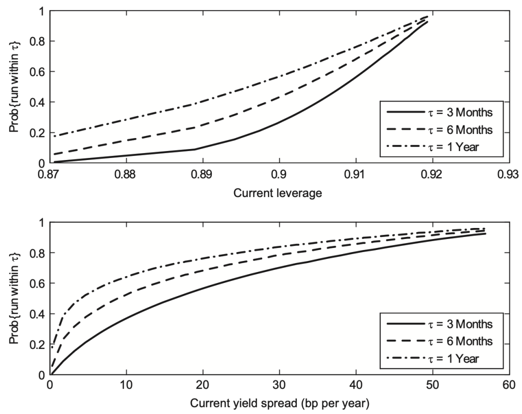

Fig. 9. The top panel plots the relation between the firm's current leverage and the probability of a run within the next three, six, and 12 months. The bottom panel shows the relation between the firm's current yield spread (basis points per year, bp/year) and the probability of a future run. Results are from model simulations using the parameter estimates for the weak-guarantee subsample, which contains the 90 ABCP conduits with SIV or extendible guarantees (Table 4).

11.12 Figure 9: warning signs

Figure 9 maps current leverage (and current spreads) into probabilities of a run within 3 months, 6 months, and 1 year. This provides an operational warning tool: high spreads/high leverage correspond to sharply rising predicted run probabilities at all horizons.

12 Short summary

1. **Endogenous yields matter**: flexible rollover rates generate dilution risk, making runs much more likely than fixed-yield rollover models in many calibrations.

2. **Structural estimation on crisis micro-data is feasible**: DTCC transaction data + a clean run definition allow separate identification of $(\theta, \sigma, \bar{r}, \alpha)$.

3. **Leverage and liquidity are first-order**: run risk is extremely nonlinear in leverage and highly sensitive to modest changes in liquidation losses.

4. **Guarantees shape recovery dynamics**: weaker guarantees increase both run incidence and post-run distress.

5. **Accounting for sponsor balance-sheet absorption is essential**: a large part of "market runs" is a shift of funding onto sponsor balance sheets (Fig. 7).

13 Conclusion

13.1 Summary

This paper concludes that the 2007 ABCP episode is well described by a **fundamental-driven, dynamic short-term debt run model** in which **rollover yields are endogenously time-varying** and therefore generate a new amplification mechanism: **dilution risk**. The key message is that **fragility is extremely nonlinear**, and in the data it is primarily governed by **leverage**

and expected liquidation costs (asset illiquidity), rather than by maturity mismatch or asset volatility.

The findings connect in one consistent chain:

1. **Empirical facts the model targets.** In the ABCP crisis, spreads rose dramatically before runs; spread volatility rose with spread levels; and high spreads forecasted near-term run risk. At the aggregate level, outstanding ABCP contracted sharply in late 2007.

2. **A structural model with flexible rollover yields matches these facts.** The model is estimated using ABCP issuance and yield data (program-by-week) and reproduces multiple salient crisis features, including the sharp contraction in ABCP outstanding (the paper reports explaining about **73%** of the decline in the second half of 2007), the rise in yields leading up to runs, the positive relation between yields and future runs, and the yield-volatility patterns.

3. **Dilution risk is a first-order theoretical amplification.** Allowing yields to rise as fundamentals deteriorate forces the conduit to promise higher face value to rolling lenders, which increases total promised debt and dilutes creditors who mature later. Anticipating this, lenders demand higher yields earlier, which accelerates leverage build-up and makes runs more likely. Quantitatively, introducing dilution risk can raise run likelihood by **more than an order of magnitude** relative to a fixed-yield benchmark.

4. **Estimated fragility is driven mainly by leverage and liquidation costs.** Structural estimation implies runs occur at very high leverage (i.e., even when institutions appear close to solvent): the predicted run threshold is approximately **92% leverage** for weak-guarantee conduits and **97.1% leverage** for strong-guarantee conduits. At the same time, the model implies that **modest improvements in liquidity** (higher recovery in liquidation) or **modest reductions in leverage** can materially reduce run probabilities.

5. **Other primitives matter less for run incidence in this setting.** The paper finds run probabilities are **much less sensitive** to the degree of maturity mismatch, the perceived strength of guarantees, and asset volatility (relative to leverage and liquidation costs), even though guarantees clearly matter for *post-run dynamics* (e.g., how long credit support lasts once a run begins). For example, the estimated "guarantee weakness" parameter implies expected survival in a run of about **262 days** (strong guarantees) versus **82 days** (weak guarantees) before backstops fail.

6. **Fit is strongest where runs are most prevalent.** The model fits substantially better in the subsample with **weak guarantees** (SIVs and extendible notes), and it is less accurate for strong-guarantee conduits when yields are high (where it tends to overpredict runs).

13.2 Why flexible rollover yields create fragility

The mechanism can be understood through three linked steps: *(i) rollover pricing* $\rightarrow$ *(ii) debt growth* $\rightarrow$ *(iii) a run threshold.*

1) Rollover requires compensation that depends on the state. Let R_t denote the promised face value per \$1 rolled over at time t. When fundamentals worsen, lenders require a higher R_t to break even.

2) Higher rollover promises mechanically increase total promised debt (dilution). Because a fraction of debt matures each instant, rolling over at face value $R_t > 1$ increases total promised debt:

$$\mathrm{d}D_t = \delta D_t(R_t - 1)\,\mathrm{d}t.$$

Thus, raising R_t to convince *today's* lenders to roll over dilutes *tomorrow's* lenders by increasing the face value they are junior to.

3) The feedback loop: higher R_t worsens leverage dynamics, which pushes R_t even higher. Define inverse leverage $x_t \equiv y_t/D_t$ (fundamental value divided by promised debt). Then

$$\frac{\mathrm{d}x_t}{x_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}Z_t - \delta(R_t - 1)\,\mathrm{d}t.$$

When R_t rises, the drift of x_t falls: inverse leverage deteriorates faster, pushing the system closer to a run. Because lenders understand this feedback, they demand higher yields earlier to compensate for the *risk of being diluted in the future* (*dilution risk*), which further accelerates the approach to the run region.

4) Why a yield cap matters. Institutionally, rollover yields cannot increase without bound, so the model imposes a cap $\bar{R}$ (equivalently, a cap on yield spreads). A run occurs when the break-even required promise exceeds the cap:

$$R_t^{\text{breakeven}}(x_t) > \bar{R} \quad \Rightarrow \quad \text{maturing creditors run.}$$

This generates a threshold x^* (a "run point") even in a purely fundamentals-driven setting.

5) Why leverage and liquidation costs dominate. Near the run point, the key object is the comparison between (i) the safe payoff from running today and (ii) the expected payoff from rolling over, which depends heavily on how much value is preserved in liquidation (asset liquidity) and on how close the institution is to solvency (leverage). Maturity mismatch, volatility, and guarantees matter, but in the paper's estimates they move outcomes less than leverage and liquidation costs.

13.3 What to remember (policy-relevant takeaways)

- **Warning signal:** high short-term yields (spreads) are an empirically and structurally meaningful early-warning indicator of runs.
- **Fragility is nonlinear:** at high leverage, very small deteriorations in asset values can trigger runs.

- **High leverage is central, yet opaque:** since leverage is so important, limited systematic access to intermediary leverage data is a serious monitoring problem.

- **Leverage and liquidity are the first-order intervention margins:** small improvements in liquidation outcomes or modest equity/leverage reductions can have outsized effects.

13.4 The paper's own suggested extensions (for future work)

The paper highlights three natural next steps:

1. endogenize the rollover-yield cap (why financing markets "refuse" to clear beyond some spread),

2. incorporate coordination failures/renegotiation in dispersed short-term debt,

3. apply the same dynamic-run estimation framework to other run-prone settings (e.g., money market funds and sovereign debt).

13.5 Takeaway

Schroth, Suarez, and Taylor (2014) show that time-varying rollover yields create dilution risk that can amplify short-term debt fragility by more than an order of magnitude, and structural estimation on ABCP data implies run risk is driven primarily by leverage and expected liquidation costs (with runs occurring at very high leverage), while maturity mismatch, guarantees, and volatility play smaller roles for run incidence in this setting.